

Q1. Hepatic Encephalopathy

Q2. What are the causes of liver abscess? Describe the clinical features of liver abscess and complications.

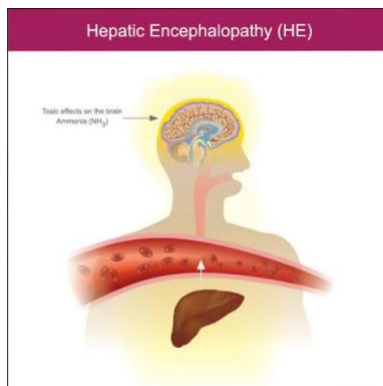
Q3. Describe clinical features, complications, diagnosis and management of serum hepatitis B. Add a note on its prevention.

Q4. Classify Cirrhosis of Liver and discuss the clinical features, diagnosis and complications of cirrhosis of liver.

Q5. Classify Jaundice. Describe its aetiology , clinical features, diagnosis, complications and management.

Q6. Define and describe clinical features of Ascites. Describe the pathogenesis and treatment of cirrhotic ascites.

Q1. Hepatic Encephalopathy



Hepatic Encephalopathy

A reversible **neuropsychiatric disorder** caused by severe liver dysfunction.

Symptoms:

- **Altered consciousness**
- **Personality changes**
- **Asterixis** (flapping tremor)
- **Coma** in severe cases

Causes:

- **Cirrhosis of the liver**

- **Acute liver failure**
- **Portosystemic shunt**
- **Infection, constipation, dehydration, or drugs** can worsen the condition

Pathophysiology:

- Accumulation of **toxins** (mainly **ammonia**) due to **impaired liver function**
- These toxins affect the brain and lead to **encephalopathy**

Diagnosis:

- **Clinical symptoms**
- **Blood tests** showing liver dysfunction
- **EEG** (electroencephalogram) may show **slowing of brain activity**

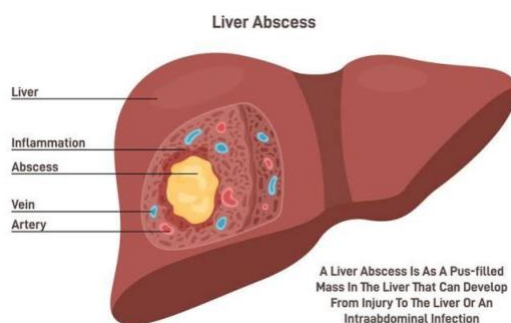
Management:

- **Lactulose** to reduce **ammonia production**
- **Antibiotics** (like **rifaximin**) to reduce gut bacteria producing toxins
- Avoid **sedatives** and **diuretics**
- Treat underlying **liver disease**

Prognosis:

- Depends on **liver function** and the cause of **encephalopathy**
- **Early treatment** can reverse symptoms
- **Severe cases** may require **liver transplantation**

Q2. What are the causes of liver abscess? Describe the clinical features of liver abscess and complications.



Bacteria reach the **liver** through:

- **Portal vein bacteremia**

- **Systemic bacteremia**
 - **Ascending cholangitis**
 - **Penetrating trauma**
- These lead to **liver abscess** formation.

Portal Vein Bacteremia

Portal vein bacteremia refers to the presence of **bacteria** in the **portal vein**, a major blood vessel that carries blood from the gastrointestinal tract to the liver. This occurs when bacteria from infections in the intestines or abdominal organs enter the bloodstream and reach the liver, causing infection.

Systemic Bacteremia

Systemic bacteremia means the spread of **bacteria** throughout the **body via the bloodstream**. It can occur due to infections in any part of the body, such as pneumonia or urinary tract infections. When these bacteria reach the liver, they may cause abscess formation.

Ascending Cholangitis

Ascending cholangitis is a bacterial infection that spreads **upwards through the bile ducts** from the small intestine. This occurs when the bile ducts are blocked (e.g., by gallstones), allowing bacteria to move toward the liver and cause infection.

Penetrating Trauma

Penetrating trauma refers to any injury that **breaks through the skin and deeper tissues**, such as a stab wound or a surgical procedure. When such injuries occur near the liver, they can introduce bacteria directly into the liver tissue, leading to abscess formation.

Common organisms involved:

- **E. coli**
- **Anaerobic streptococci**
- **Bacteroides**

Most cases have a **subacute onset**.

Clinical symptoms:

- **Fever with chills and rigors**
- **Weight loss**
- **Anorexia**
- **Nausea and vomiting**

Other signs:

- **Pleuritic chest pain**
- **Tender hepatomegaly**
- **Mild jaundice**

Single abscesses:

- More common in the **right lobe** of the **liver**
- **Multiple abscesses:** Common in **elderly patients**

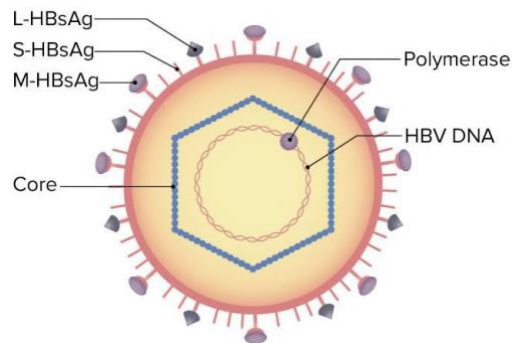
Treatment:

- Start with **ampicillin, gentamicin, and metronidazole**.
- Change the **antibiotic** later based on **sensitivity reports**.

Procedures:

- **Ultrasound-guided aspiration** of the **abscess**
- **Surgical drainage** when necessary

Q3. Describe clinical features, complications, diagnosis and management of serum hepatitis
B. Add a note on its prevention.



Viruses causing Acute Viral Hepatitis:
Specific hepatitis viruses:

- **Hepatitis A virus (HAV)**
- **Hepatitis B virus (HBV)**
- **Hepatitis C virus (HCV)**
- **Hepatitis D virus (HDV)**
- **Hepatitis E virus (HEV)**

Other viruses:

- **Cytomegalovirus**
- **Epstein-Barr virus**
- **Herpes simplex virus**

Hepatitis B

- **DNA virus**
- **Core** contains DNA and DNA polymerase enzyme for virus replication
- **Humans** are the only source of infection
- Transmitted by **infected blood, blood products, sexual contact**, and **from infected mother to child during delivery**
- **Tattooing** and **acupuncture** can also spread it

Route of Transmission

- **Horizontal transmission:**
 - Injection drug use
 - Infected unscreened blood
 - Tattooing, acupuncture needles
 - **Sexual** (homosexual and heterosexual)
- **Vertical transmission:**
 - **HbsAg-positive mother**

High-Risk Groups:

- **Haemodialysis** patients, **physicians, surgeons, dentists, paramedical staff**, and **laboratory/blood bank workers**
- Infected individuals are contagious during **incubation period** and as long as **HBsAg-positive**

Acute Hepatitis Clinical Features

1. Prodromal phase:

- **Anorexia, nausea, vomiting, headache, fatigue, malaise, myalgia, arthralgia**
- **Low-grade fever**

2. Icteric phase:

- **Jaundice** appears with dark urine and yellowish discoloration of skin and eyes
- **Clay-colored stool** and **pruritus** (suggests **cholestasis**)
- Some may never get jaundice (anicteric hepatitis)

3. Recovery phase:

- Improvement in symptoms and reduction in jaundice
- **Clinical and biochemical recovery** within 3-4 months

Signs of Acute Hepatitis:

- **Yellow sclera**
- **Skin scratch marks** due to **pruritus**
- **Tender hepatomegaly** in more than 50% cases
- **Splenomegaly** and **lymphadenopathy** in 10-20% cases

Complications of Acute Hepatitis:

- **Acute fulminant hepatitis**
- **Relapsing hepatitis**
- **Cholestatic hepatitis**
- **Chronic hepatitis** and **cirrhosis**
- **Aplastic anaemia**
- **Papular acrodermatitis**
- **Myelitis** and **neuropathy**

Investigations

Serology

- **HBV** has **3 important antigens**:
 - **HBsAg** (Hepatitis B Surface Antigen)
 - **HBcAg** (Core Antigen)
 - **HBeAg** (Hepatitis E Antigen)
- **HBsAg**:
 - Appears first in serum as evidence of infection
 - Persists for **3-4 weeks**, but can last up to **6 months**
 - After disappearance, **Anti-HBs** (antibody to HBsAg) appears and **confers immunity**
 - **Anti-HBs** means either **previous infection** or **vaccination**
- **HBcAg**:
 - Not found in blood
 - **Anti-HBc** appears early during illness
 - **IgM anti-HBc** indicates **acute infection**
 - **IgG anti-HBc** suggests **chronic infection** (when HBsAg is positive) or **recovery** (when Anti-HBs is positive)
- **HBeAg**:
 - Indicates **active viral replication** and high infectivity

- **Anti-HBe** appears when HBeAg disappears, indicating **low viral replication** and decreased infectivity

Viral Blood

- **HBV-DNA** can be measured by **PCR** in the blood
- **Viral load > 105 copies/ml** indicates **active viral replication**

Other Investigations

- Significant increase in **ALT** and **AST** (>400 units/L) during early hepatitis
- **High alkaline phosphatase** suggests **cholestasis**
- **Prolongation of prothrombin time (PT)** indicates severe liver damage

Management

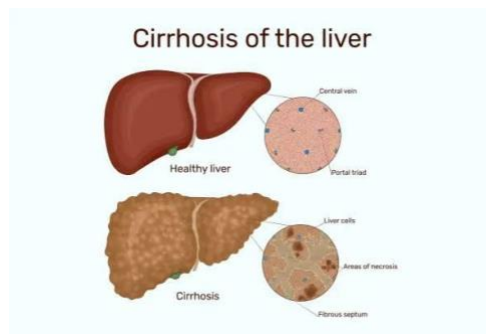
Acute Hepatitis

- **Supportive treatment** and monitoring for **acute liver failure**
 - **Physical activity** should be restricted, **bed rest** in severe cases
 - **High-calorie diet** with good **protein intake**
 - **Hospitalization** and **IV fluids** (10% glucose) if oral intake is inadequate or severe nausea/vomiting
 - Avoid **hepatotoxic drugs** or those metabolized in the liver
 - **Cholestyramine** for **pruritus** in cases with **cholestasis**
 - **Severe hepatic failure** (altered mental status, prolonged PT/bleeding) requires **hospitalization**
 - No specific therapy for **acute viral hepatitis**, except for **acute HCV infection** (treat with **subcutaneous interferon alpha**)
 - **Liver transplantation** may be required in **acute fulminant hepatic failure**

Prevention of Viral Hepatitis

- Prophylaxis available for **HAV** and **HBV**
- **Active immunization** (pre-exposure prophylaxis):
 - **Recombinant hepatitis B vaccine** containing **HBsAg**
 - **95% efficacy** in producing immunity in normal individuals

Q4. Classify Cirrhosis of Liver and discuss the clinical features, diagnosis and complications of cirrhosis of liver.



- **End result of hepatocellular injury**
- Characterized by **extensive fibrosis, regenerative nodules, and loss of liver architecture**

Aetiology

1. **Common causes:**
 - **Viral hepatitis (B, C, D)**
 - **Prolonged excessive alcohol use**
2. Other important causes:
 - **Autoimmune hepatitis**
 - **Drug-induced**
 - **Biliary cirrhosis**
 - **Hemochromatosis**
 - **Wilson's disease**
 - **Cardiac cirrhosis**
 - **Alpha-1 antitrypsin deficiency**
 - **Idiopathic**

Pathogenesis

- **Activation of stellate cells** is central to cirrhosis development
- Stellate cells transform into cells that produce **type I collagen**, leading to fibrosis
- **Micronodular cirrhosis** (small regenerating nodules) is common in alcoholics
- **Macronodular cirrhosis** (larger nodules) is seen in post-hepatic or post-necrotic cirrhosis

Clinical Features

- **Asymptomatic** in some cases, but can present with:
 - **Weakness, fatigue, weight loss, anorexia, nausea, vomiting, abdominal discomfort**
- **Portal hypertension:**

- **Ascites, splenomegaly, haematemesis, melaena, hepatic encephalopathy**
- **Liver size** may reduce as disease progresses due to fibrosis
- **Hepatocellular dysfunction** causes:
 - **Jaundice, oedema, coagulopathy, metabolic abnormalities**
 - **Spider nevi, palmar erythema, gynaecomastia, testicular atrophy, hair loss**
 - **Females** may have **loss of libido, menstrual abnormalities, breast atrophy**
 - **Spider nevi** (dilated arterioles) mostly on upper body
- Other signs:
 - **Ascites, haemorrhagic manifestations** (epistaxis, increased menstrual flow), **parotid gland enlargement, digital clubbing, Dupuytren's contracture, skin pigmentation**

Complications

1. **Portal hypertension**
2. **Ascites**
3. **Upper GI bleeding**
4. **Spontaneous bacterial peritonitis (SBP)**
5. **Hepatic encephalopathy**
6. **Hepatorenal syndrome**
7. **Hepatocellular carcinoma**

Laboratory Findings

1. **Blood examination:**
 - **Anaemia, leukopenia, thrombocytopenia**
 - **Elevated ALT/AST, bilirubin, and ALP**
 - **Low serum albumin and prolonged PT**
2. **Imaging:**
 - **Ultrasonography** to check liver size, texture, ascites, portal hypertension, splenomegaly
3. **Endoscopy:**
 - Detects **oesophageal varices**, excludes other upper GI bleeding causes
4. **Liver biopsy:**
 - Assesses cirrhosis severity and identifies the specific cause

Management

1. **General management:**
 - Adequate **protein** and **calories** in diet
 - **Vitamin supplementation**
 - **Salt restriction** in ascites

- Avoid **hepatotoxic drugs**
- 2. **Treatment of specific causes:**
 - **Alcohol abstinence** in alcoholic cirrhosis
 - Specific therapy for **hemochromatosis** and **Wilson's disease**
- 3. **Management of complications**
- 4. **Liver transplantation:**
 - Most common indication for **irreversible chronic liver failure**

Prognosis

- **Poor prognosis**, but **favorable if the cause is corrected**

Q5. Classify Jaundice. Describe its aetiology , clinical features, diagnosis, complications and management.



- **Yellowish pigmentation** of skin, mucous membranes, and sclera due to elevated bilirubin levels in the blood.
- Scleral pigmentation occurs because sclera has a high amount of **elastin**, which binds to bilirubin.
- Clinical jaundice appears when **serum bilirubin** is greater than **3 mg%** (normal: 0.3 - 1.0 mg%).
- **Latent jaundice** occurs when serum bilirubin is between **1 - 3 mg%**.
- **Clinical detection** is easier in daylight than artificial light.
- **Carotenaemia** can also cause yellow skin but does not affect the sclera.

Classification of Jaundice

Jaundice can be classified based on:

1. **Underlying bilirubin metabolism derangement**
2. **Pathological mechanism**

A. Classification Based on Bilirubin Metabolism

1. **Predominantly Unconjugated Hyperbilirubinaemia**
 - **Overproduction of bilirubin**
 - **Decreased hepatic uptake**
 - **Decreased conjugation of bilirubin**
2. **Predominantly Conjugated Hyperbilirubinaemia**
 - **Intrahepatic causes**
 - **Extrahepatic causes**

Unconjugated Hyperbilirubinaemia

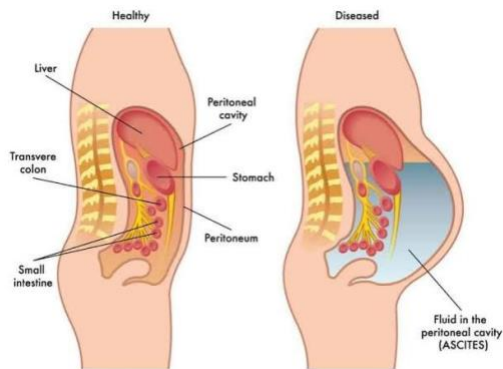
- Characterized by high **unconjugated bilirubin** levels in serum.
 - **Urine** color is **normal**.
1. **Overproduction of Bilirubin (Prehepatic)**
 - Occurs in **hemolytic disorders** like:
 - **Hereditary spherocytosis, sickle cell anaemia, G6PD deficiency, paroxysmal nocturnal haemoglobinuria, autoimmune haemolytic anaemia**
 - Other causes: ineffective erythropoiesis, haemolytic reactions, resolution of hematomas.
 2. **Impaired Hepatic Metabolism**
 - Impaired **hepatic uptake** or **conjugation** of bilirubin:
 - **Gilbert's syndrome** (autosomal dominant, mild jaundice)
 - **Crigler-Najjar syndrome** (severe jaundice, type I has complete enzyme deficiency, type II has reduced enzyme activity)
 - **Neonates** may have mild unconjugated bilirubin rise (physiologic neonatal jaundice). **Premature infants** with severe elevation can get **kernicterus**.

Conjugated Hyperbilirubinaemia

- **Serum conjugated bilirubin** is elevated.
1. **Dubin-Johnson and Rotor Syndromes**
 - Rare inherited disorders causing **asymptomatic jaundice**.
 - **Normal liver enzymes**.
 2. **Hepatocellular Diseases**
 - Caused by the liver's inability to **transport bilirubin** into bile due to liver damage.
 - Causes include:
 - **Viral hepatitis** (A, B, C, D, E, EBV, CMV)
 - **Alcohol**
 - **Drugs** (e.g., isoniazid, paracetamol)

- **Toxins** (e.g., vinyl chloride, amanita phalloides)
 - **Metabolic** (e.g., Wilson's disease)
 - **Immune** (e.g., autoimmune hepatitis)
3. **Cholestatic Jaundice (Obstructive Jaundice)**
- Caused by **obstruction** in bile flow (intrahepatic or extrahepatic).
 - **Intrahepatic causes:**
 - **Hepatitis, primary biliary cirrhosis, drugs** (e.g., chlorpromazine, anabolic steroids, contraceptives)
 - **Extrahepatic causes:**
 - **Stone in bile duct, sclerosing cholangitis, neoplasms** (e.g., gallbladder, pancreas), **enlarged lymph nodes compressing ducts**
- **Conjugated bilirubin** enters blood and is excreted in **urine**.
 - **Pale stools** due to reduced stercobilinogen formation.
 - **Pruritus (itching)** due to bile salt deposition in skin.
 - **Malabsorption of fats and fat-soluble vitamins (A, D, E, K)** leading to:
 - **Bleeding diathesis** (due to vitamin K deficiency)
 - **Osteomalacia** (due to vitamin D deficiency)
 - **Elevated alkaline phosphatase** compared to aminotransferases.

Q6. Define and describe clinical features of Ascites. Describe the pathogenesis and treatment of cirrhotic ascites.



- **Definition:** Accumulation of excess fluid in the **peritoneal cavity**.

Aetiology

- **Normal Peritoneum (Transudates)**
 - **Cirrhosis** with portal hypertension
 - **Congestive heart failure (CHF)**
 - **Hypoproteinaemia** (nephrotic syndrome, malnutrition)
 - **Pancreatitis**

- **Biliary ascites**
- **Diseased Peritoneum (Exudates)**
 - **Tuberculous peritonitis**
 - **Bacterial peritonitis**
 - **Malignancy** (peritoneal, hepatic)
- **Miscellaneous**
 - **Chylous ascites** (filariasis, trauma, tumour)
 - **Meig's syndrome**
 - **Budd-Chiari syndrome**

Mechanisms of Formation

- **Inflammation of peritoneum** → increased capillary permeability and reduced fluid reabsorption (e.g., bacterial/tuberculous peritonitis).
- **Venous obstruction** (e.g., inferior vena cava obstruction) → fluid transudation into the cavity.
- **Lymphatic obstruction** → **chylous ascites** (e.g., mesenteric lymph node, thoracic duct obstruction).
- **Rupture of viscus** → leakage of blood, cystic fluid, or contaminants (e.g., pancreatitis).

Clinical Features

- **Symptoms:** Abdominal distension (with/without pain), **dyspnoea**, **orthopnoea**, indigestion, and heartburns due to increased abdominal pressure.
- **Signs:** Abdominal distension, **fullness in flanks**, **everted umbilicus**, striae, prominent veins, and hernias.
 - **Fluid thrill** → Massive ascites
 - **Shifting dullness** → Moderate ascites (needs at least 1,000 ml of fluid)
 - **Spider nevi, palmar erythema, gynaecomastia** → Chronic liver disease
 - **Splenomegaly** → Portal hypertension
 - Resonance in **epigastrium and umbilical region** (floating intestines).

Secondary Effects of Ascites

- **Scrotal oedema**
- **Pleural effusion** (especially right-sided)
- **Oedema** due to hypoproteinaemia and vena cava blockage
- **Cardiac apex shift** due to raised diaphragm
- **Neck vein distension** due to increased right atrial pressure
- **Meralgia paraesthetica** due to compression of lateral cutaneous nerve

Investigations

- **Ultrasonography:** Detects small amounts of fluid and identifies the cause.

- **Diagnostic paracentesis**
- **Laparoscopy and peritoneal biopsy**
- **Ascitic fluid examination**

Management

- **Salt restriction:** 2-3 g/day
- **Diuretics:**
 - **Spironolactone:** Start with 100 mg/day, increase to 400 mg/day.
 - **Furosemide:** 40-160 mg/day if needed.
 - Goal: Weight loss <1 kg/day in anasarca; <0.5 kg/day in ascites alone.
- **Therapeutic paracentesis:** Remove 3-5 liters in massive ascites. Give **albumin IV** to maintain volume.
- **Surgical options:**
 - **Portacaval shunt** or **peritoneovenous shunt** in refractory ascites.
 - **TIPSS (transjugular intrahepatic portosystemic stent shunt)** for resistant ascites.